

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459961.7095 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.044
Transit depth (Rp/Rs)^2	6e-05 +/- 0.0068 [%]
Orbital Inclination (inc)	78.88 +/- 2.87
Airmass coefficient 1 (a1)	0.55 +/- 0.3
Airmass coefficient 2 (a2)	0.47 +/- 0.63
Scatter in the residuals of the lightcurve fit is	51.44 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	3.75
Transit Duration (day)	0.098 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.001 ± 0.044 vs 0.1178 (deviation 2.65σ)
Duration fit vs published	0.098 d vs 0.125 d (deviation 1.8σ)
Mid-time O-C	1.4 min
Depth SNR	0.0
Residual scatter	51.44 %

Light curve

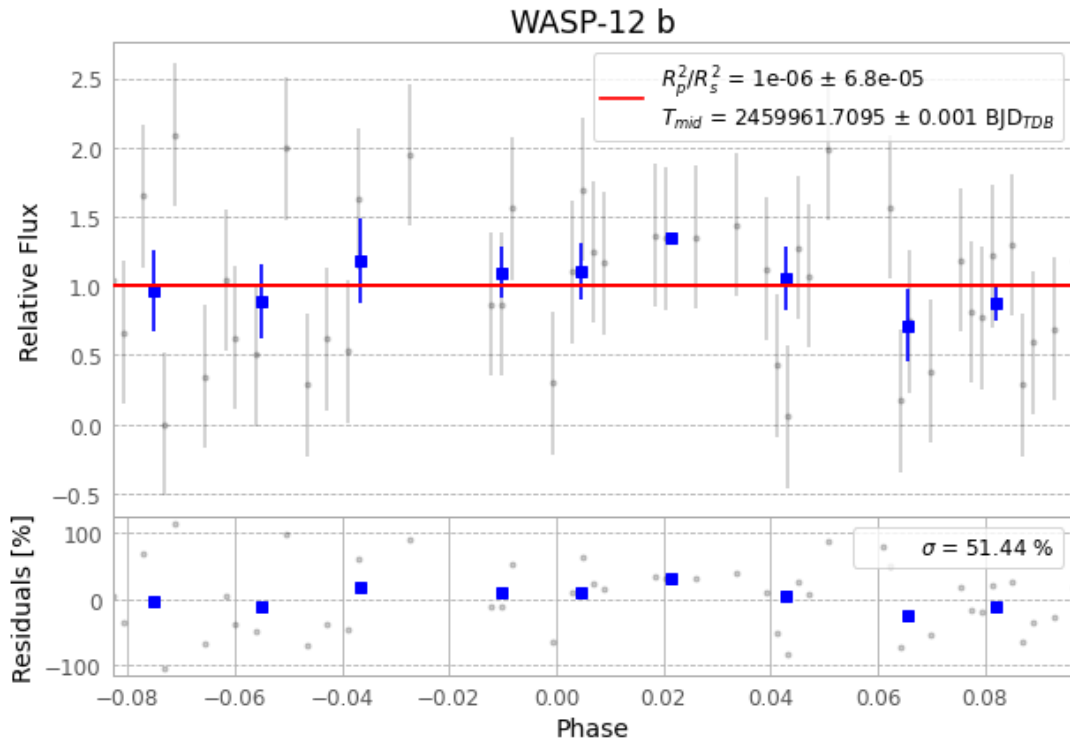


Figure 1 — FinalLightCurve_WASP-12 b_2023-01-16.png (SHA-256 06f02728a4c28a32...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [122, 446], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1f32f7bfd42bfbce...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

98 FITS frames (64 MB), WASP-12230117023615.FITS → WASP-12230117072716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230117.log (678e4ff5a33a...), FinalLightCurve_WASP-12 b_2023-01-16.png (06f02728a4c2...), FinalLightCurve_WASP-12 b_2023-01-16.csv (8a1a9e7785c5...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 03:53Z.